



University  
of Glasgow

# Power Electronics

## DC-AC Converters (Inverters)

## 逆变器

Dr Atif Jafri

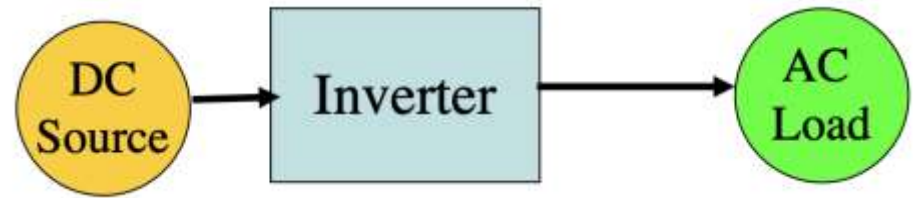
Atif.Jafri@glasgow.ac.uk

Please read **Chapter 8** in the textbook

# Scheme of Study

- Fundamental
- Types of Inverters
- Half-bridge Inverter
  - Ideal Circuit Diagram
  - Operation
  - Output Waveform
  - Harmonics
  - Practical Circuit Diagram
  - Avoiding shoot through

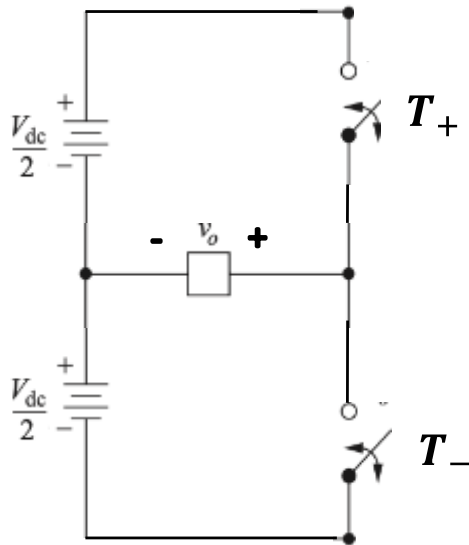
# Fundamentals



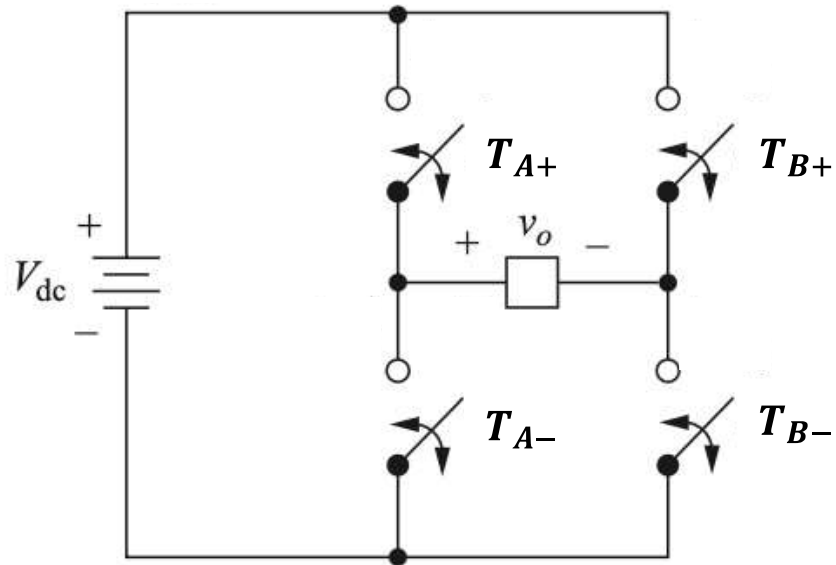
- DC-AC Converters are known as inverters
- Role is to convert a DC signal to AC
- Ideally, output should be sinusoidal.
- In reality, these are non-sinusoidal and contain harmonics
- This is fine for low and medium power applications
- Divided into two main types
  - Single Phase
  - Three Phase
- Semiconductor devices typically used
- The basic building block is the 'Bridge' circuit

# Inverter Types

## - Half and Full Bridge



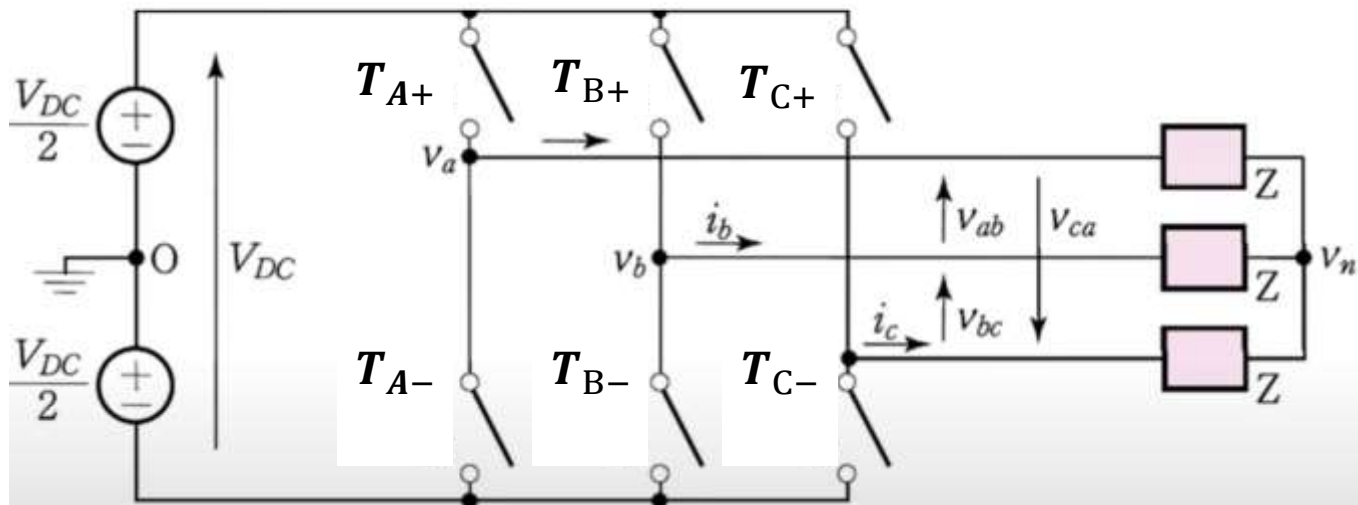
Half Bridge Inverter (Ideal)



Full Bridge Inverter (Ideal)

# Inverter Types

## 3-Leg (For 3-phase inverter)



3-Leg Inverter (Ideal)

# Schemes of Inverters to study

| Phase Output | Bridge | Switching Schemes                           | <ul style="list-style-type: none"><li>• Ideal Circuit Diagram</li><li>• Operation</li><li>• Output Waveform</li><li>• Harmonics</li><li>• Practical Circuit Diagram<ul style="list-style-type: none"><li>• Avoiding shoot through</li></ul></li></ul> |
|--------------|--------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single Phase | Half   | Square Wave                                 |                                                                                                                                                                                                                                                       |
|              |        | Pules Width Modulation                      |                                                                                                                                                                                                                                                       |
|              | Full   | Square Wave                                 |                                                                                                                                                                                                                                                       |
|              |        | Pules Width Modulation (Bipolar & Unipolar) |                                                                                                                                                                                                                                                       |
| Three Phase  | 3-Leg  | Square Wave                                 |                                                                                                                                                                                                                                                       |
|              |        | Pules Width Modulation                      |                                                                                                                                                                                                                                                       |

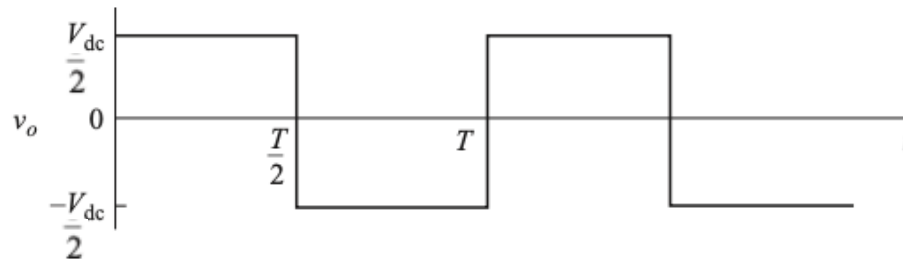
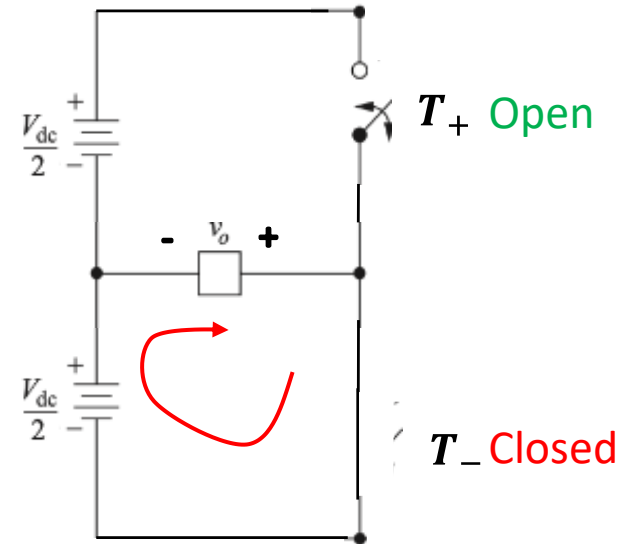
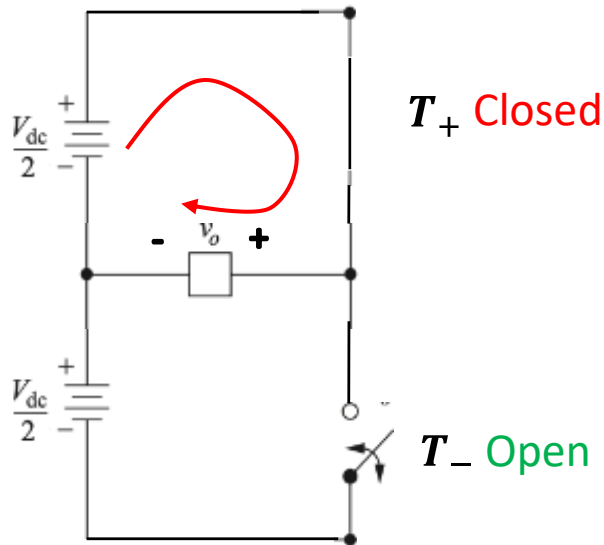
# Schemes of Inverters to study

| Phase Output | Bridge | Switching Schemes                           | <ul style="list-style-type: none"><li>• Operation in different switching schemes</li><li>• Output Waveform</li><li>• Harmonics</li><li>• Practical Circuit</li><li>• Avoiding shoot through</li></ul> |
|--------------|--------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single Phase | Half   | Square Wave                                 |                                                                                                                                                                                                       |
|              |        | Pules Width Modulation                      |                                                                                                                                                                                                       |
|              | Full   | Square Wave                                 |                                                                                                                                                                                                       |
|              |        | Pules Width Modulation (Bipolar & Unipolar) |                                                                                                                                                                                                       |
| Three Phase  | 3-Leg  | Square Wave                                 |                                                                                                                                                                                                       |
|              |        | Pules Width Modulation                      |                                                                                                                                                                                                       |

# Operation

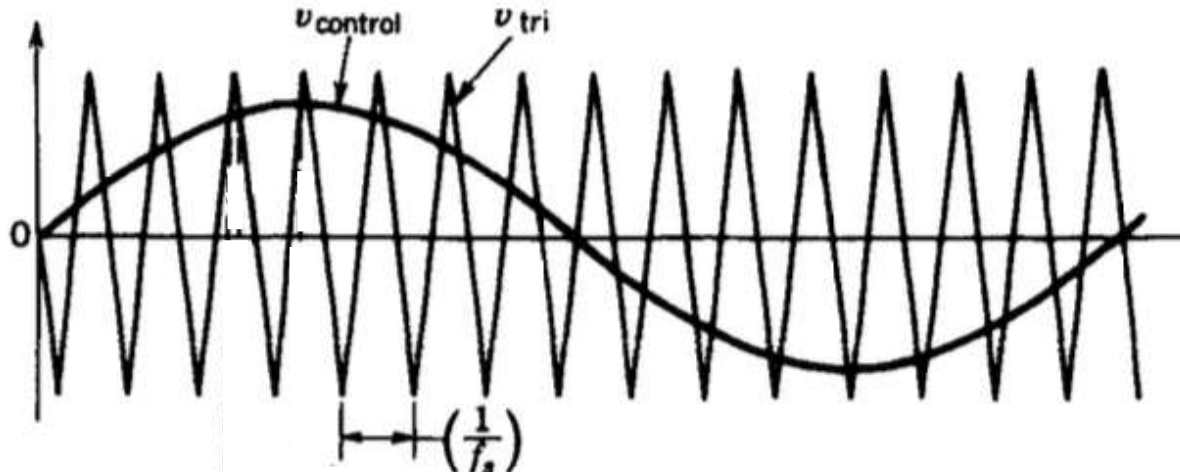
## - Square Wave Switching

Half Bridge Inverter (Ideal)



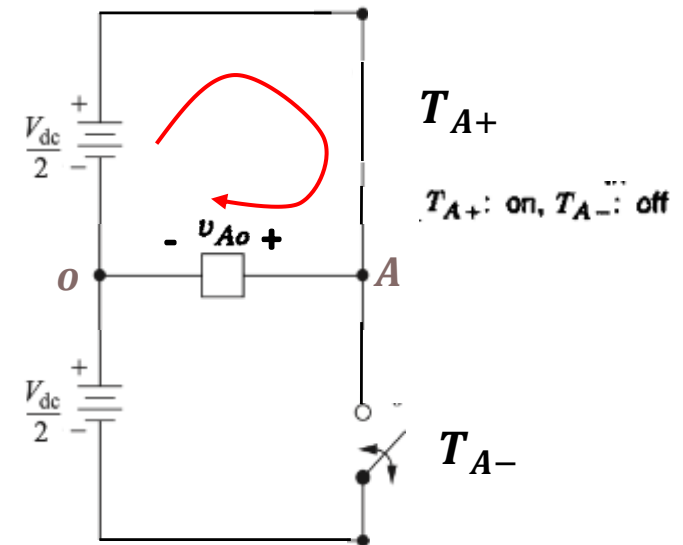
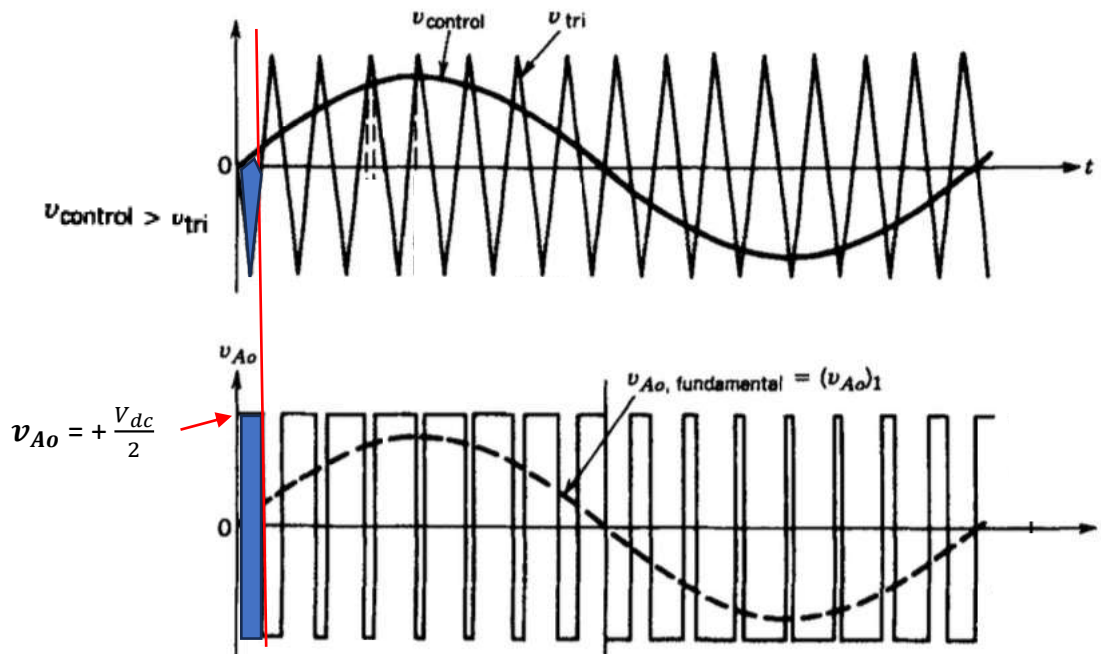
# Operation

- Pulse Width Modulation (PWM) Switching
  - Is a type of switching technique.
  - Use two signals called
    - $v_{tri}$  (Triangular Signal)
      - Has frequency  $f_s$  (called switching frequency)
    - $v_{control}$  (Control Signal)
      - Has frequency  $f_1$  (desired fundamental frequency)



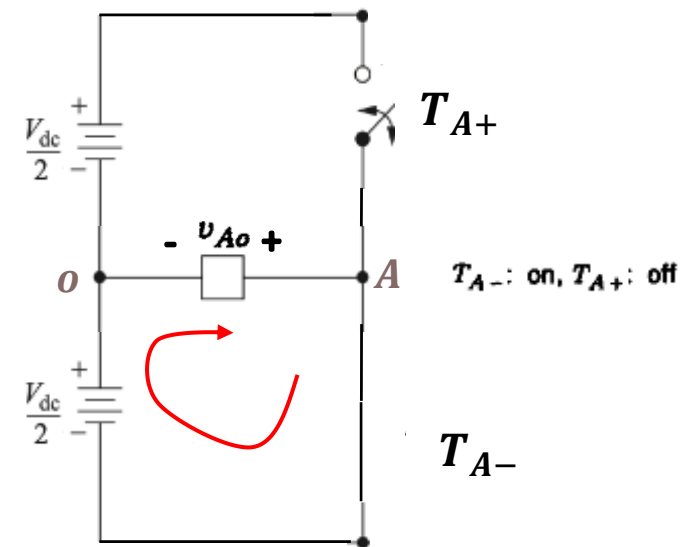
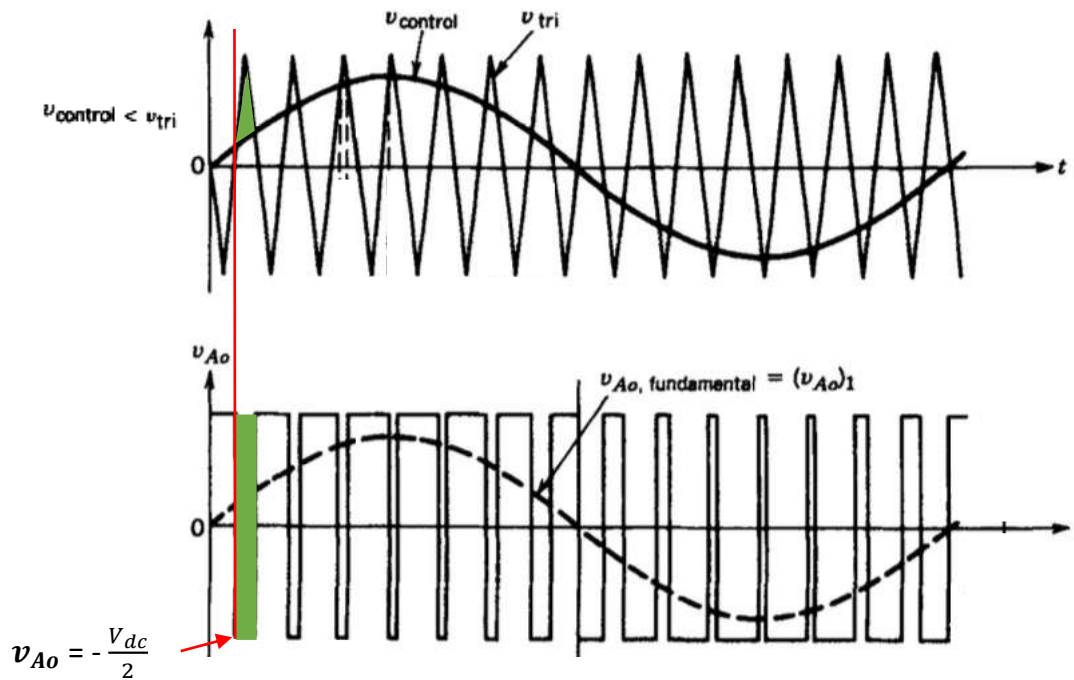
# Operation

- Pulse Width Modulation (PWM) Switching
  - $v_{tri}$  (Triangular Signal) is compared with  $v_{control}$  (Control Signal)
  - Based on comparison result different switches are made closed (ON) and open (OFF)
  - Output voltage swings between positive and negative voltages



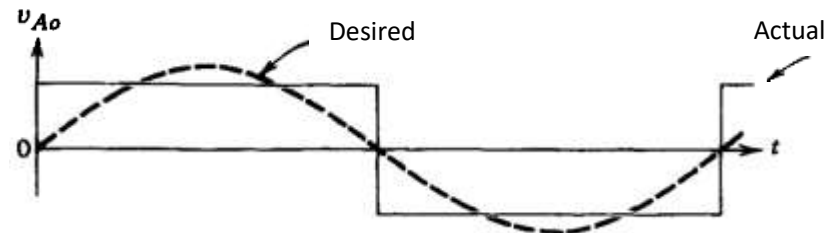
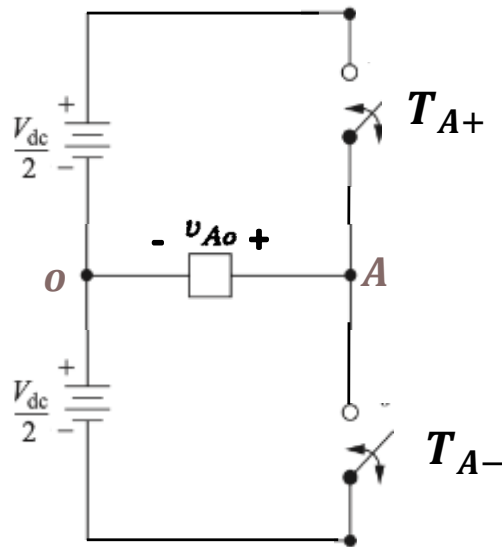
# Operation

- Pulse Width Modulation (PWM) Switching
  - $v_{tri}$  (Triangular Signal) is compared with  $v_{control}$  (Control Signal)
  - Based on comparison result different switches are made closed (ON) and open (OFF)
  - Output voltage swings between positive and negative voltages

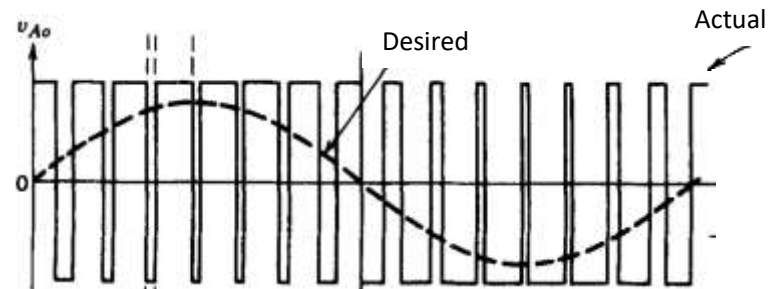


# Harmonics

- What are harmonics?
  - Pure Sinusoid is required.
  - The output is different both in square-wave and PWM switching
  - There is need to analyze the **presence of other sinusoids** having **frequencies multiple of desired sinusoid**, **called harmonics**, available in output.



Output (Square Wave Switching)

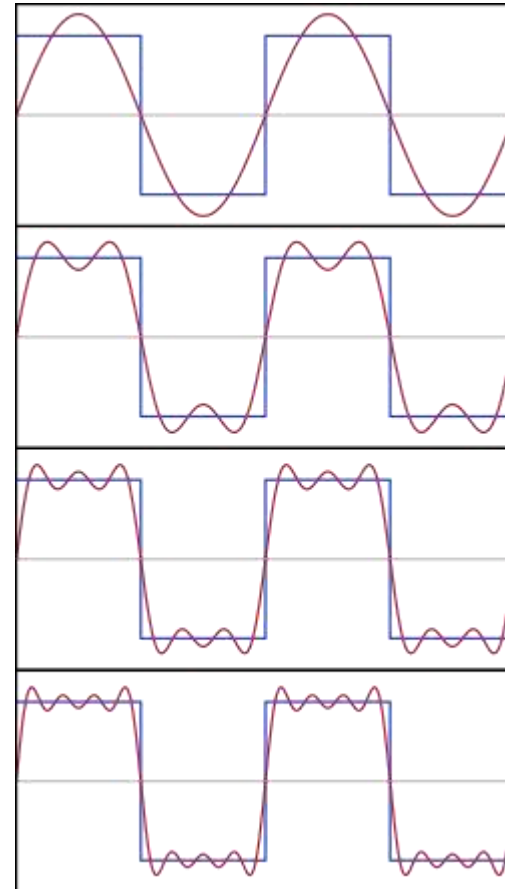
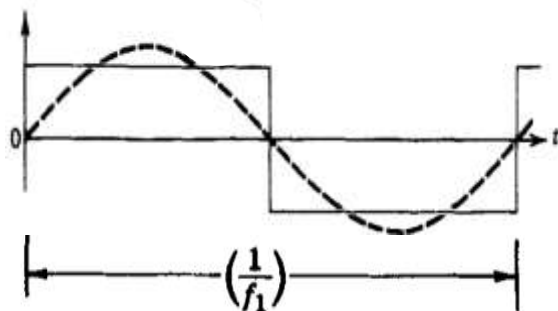
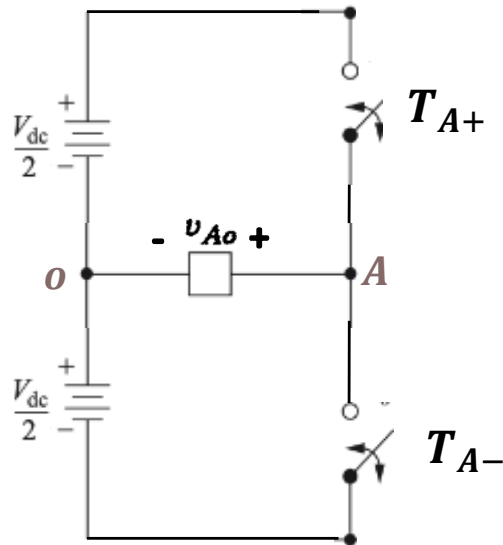


Output (PWM Switching)

# Harmonics

## - Harmonics in Square Wave Switching

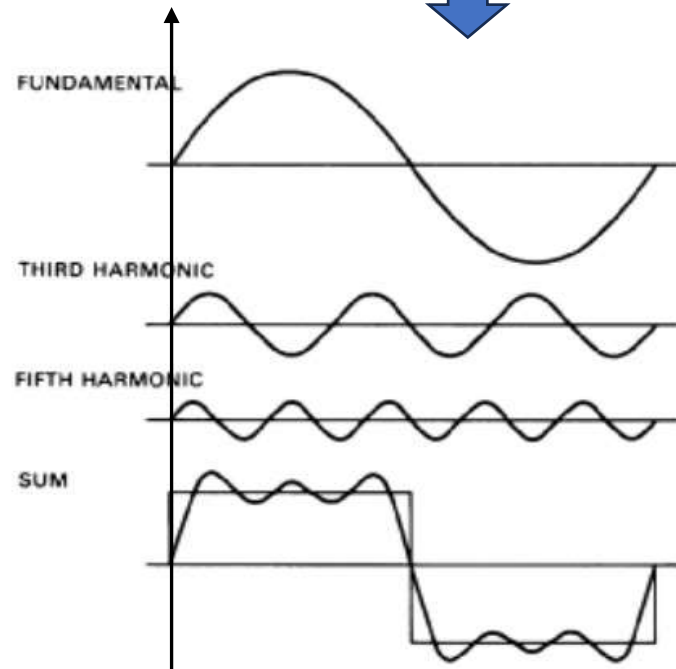
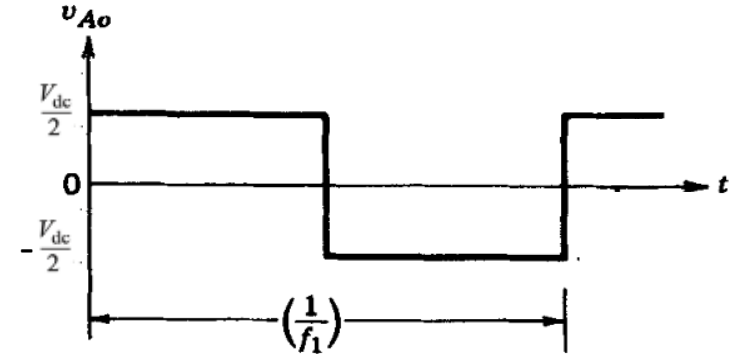
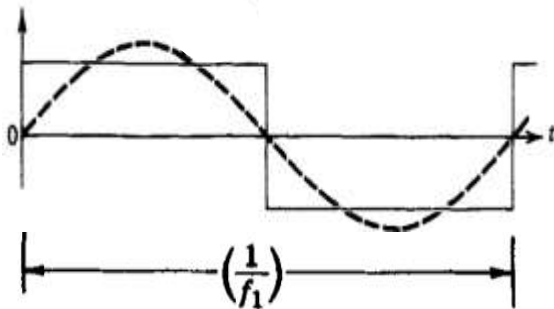
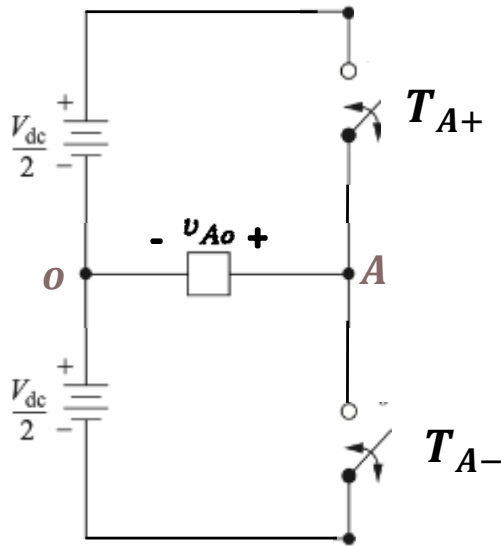
- There is need to analyze the **presence of other sinusoids** having **frequencies multiple of desired sinusoid**, **called harmonics**, available in output.



# Harmonics

## - Harmonics in Square Wave Switching

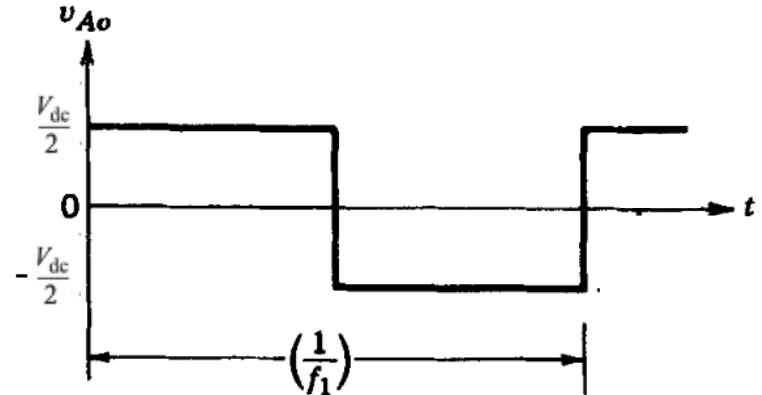
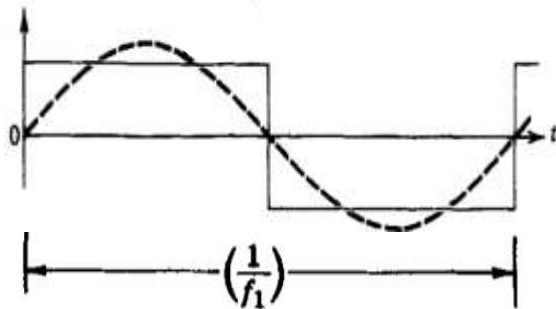
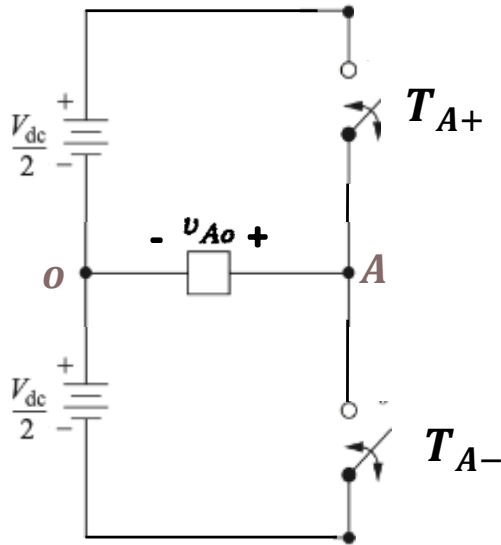
- There is need to analyze the **presence of other sinusoids** having **frequencies multiple of desired sinusoid, called harmonics**, available in output.



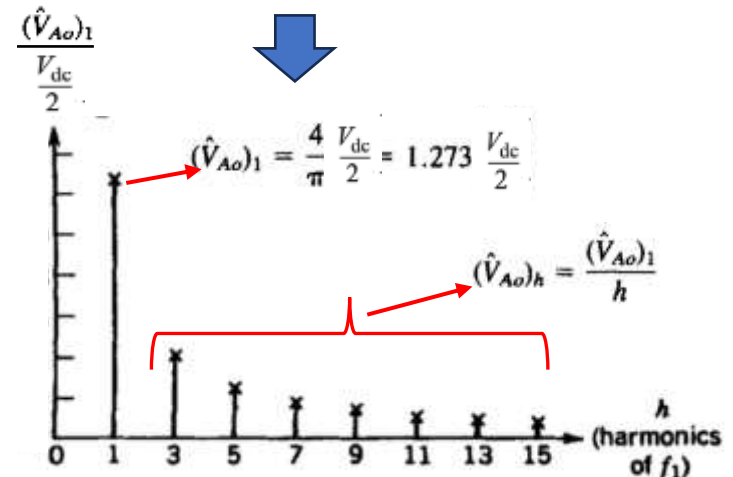
# Harmonics

## - Harmonics in Square Wave Switching

- There is need to analyze the **presence of other sinusoids** having **frequencies multiple of desired sinusoid, called harmonics**, available in output.



Fourier Series



# Harmonics

## - Harmonics in PWM Switching

- To analyze “**harmonics**” for PWM switching we need to understand two new terms:
  - **Amplitude Modulation Ratio** and **Frequency Modulation Ratio**.

$$m_a = \frac{\hat{V}_{\text{control}}}{\hat{V}_{\text{tri}}}$$



Amplitude Modulation Ratio

$$\hat{V}_{\text{control}}$$

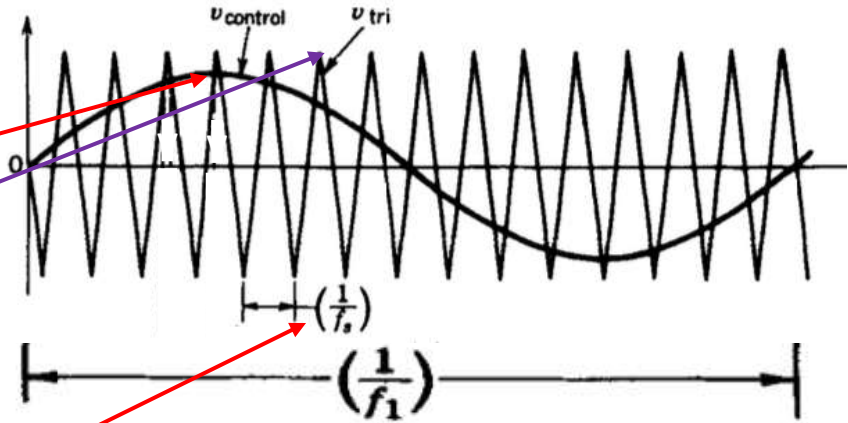


Peak Value of Control Signal

$$\hat{V}_{\text{tri}}$$



Peak Value of Triangular Signal



$$m_f = \frac{f_s}{f_1}$$



Frequency Modulation Ratio

$$f_s$$



Frequency of triangular signal  
also called switching frequency

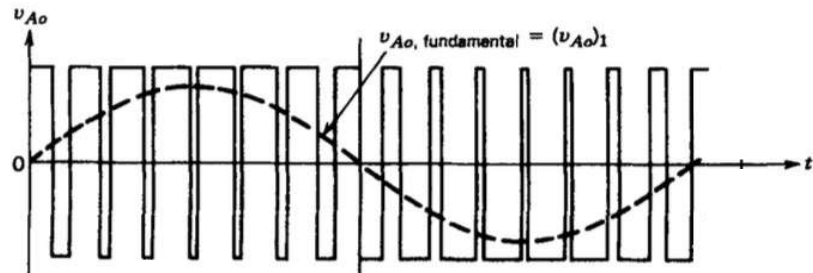
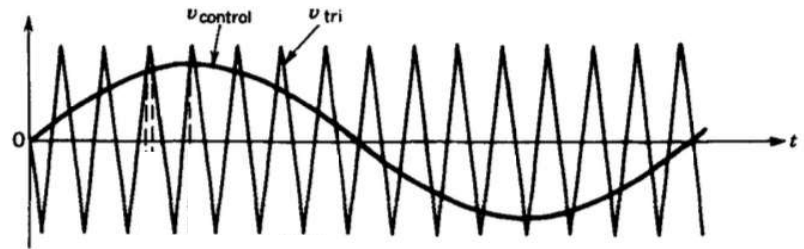
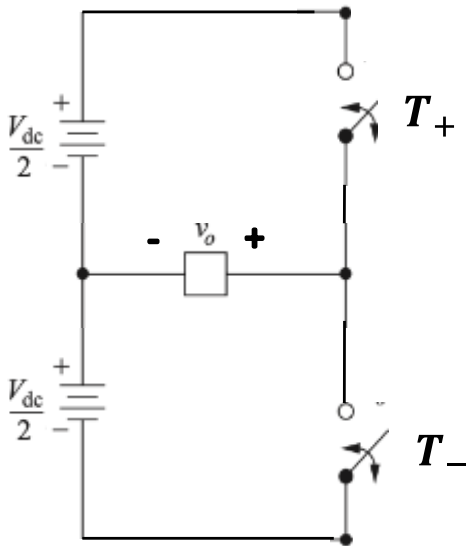
$$f_1$$



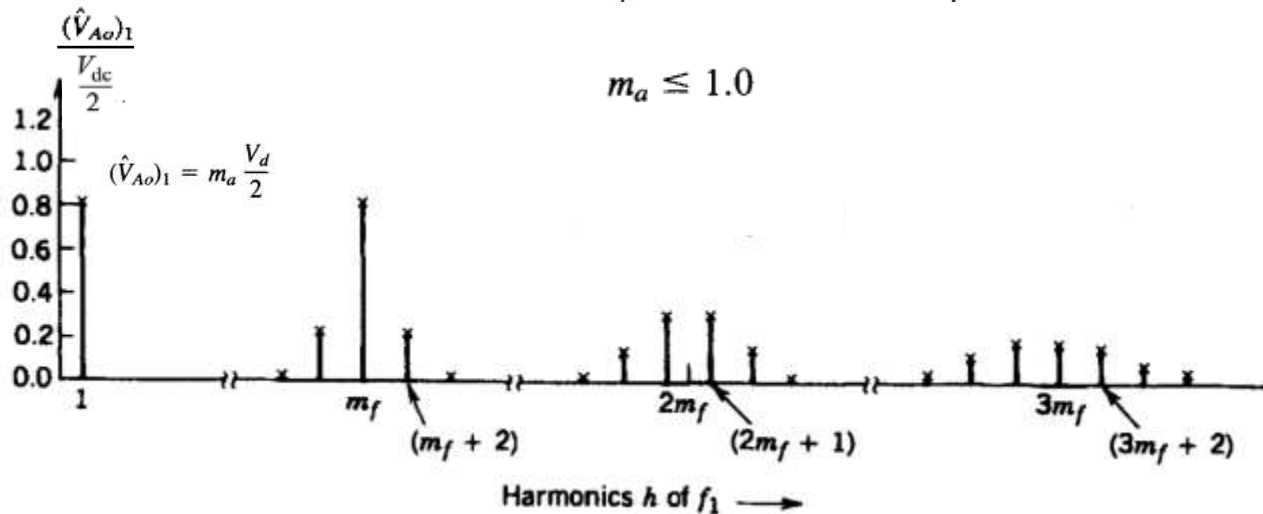
Frequency of control signal also  
called desired fundamental frequency

# Harmonics

## - Harmonics in PWM Switching

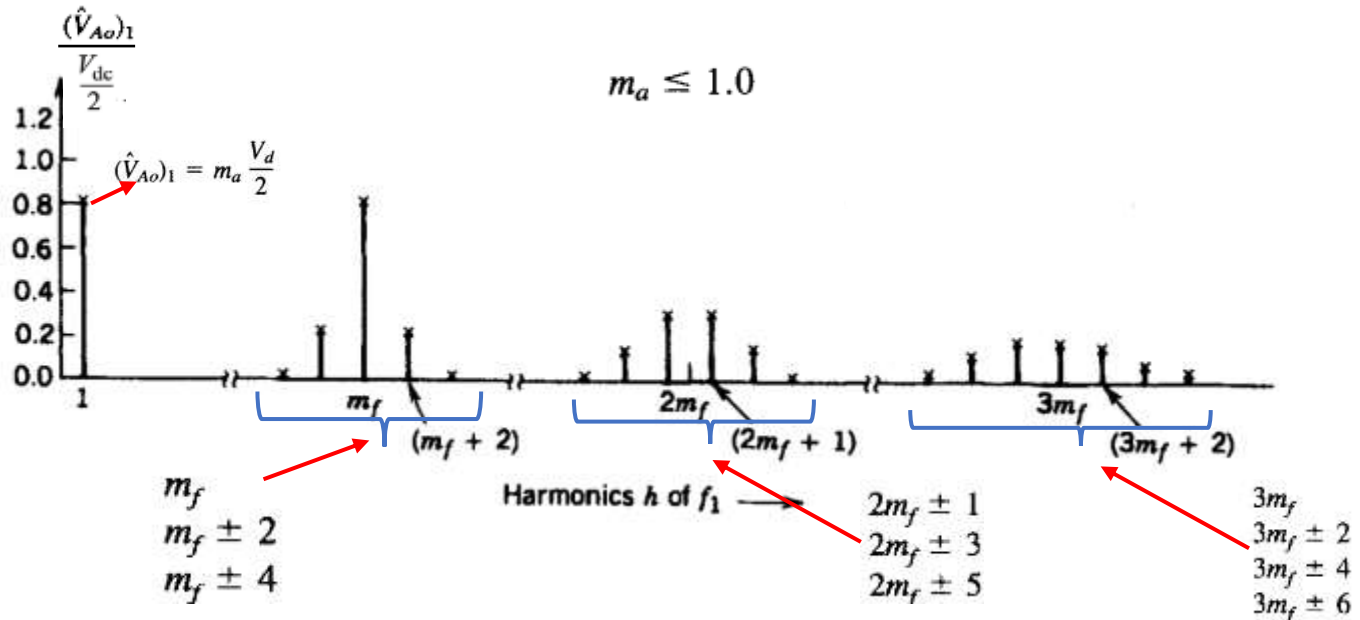


$$m_a \leq 1.0$$



# Harmonics

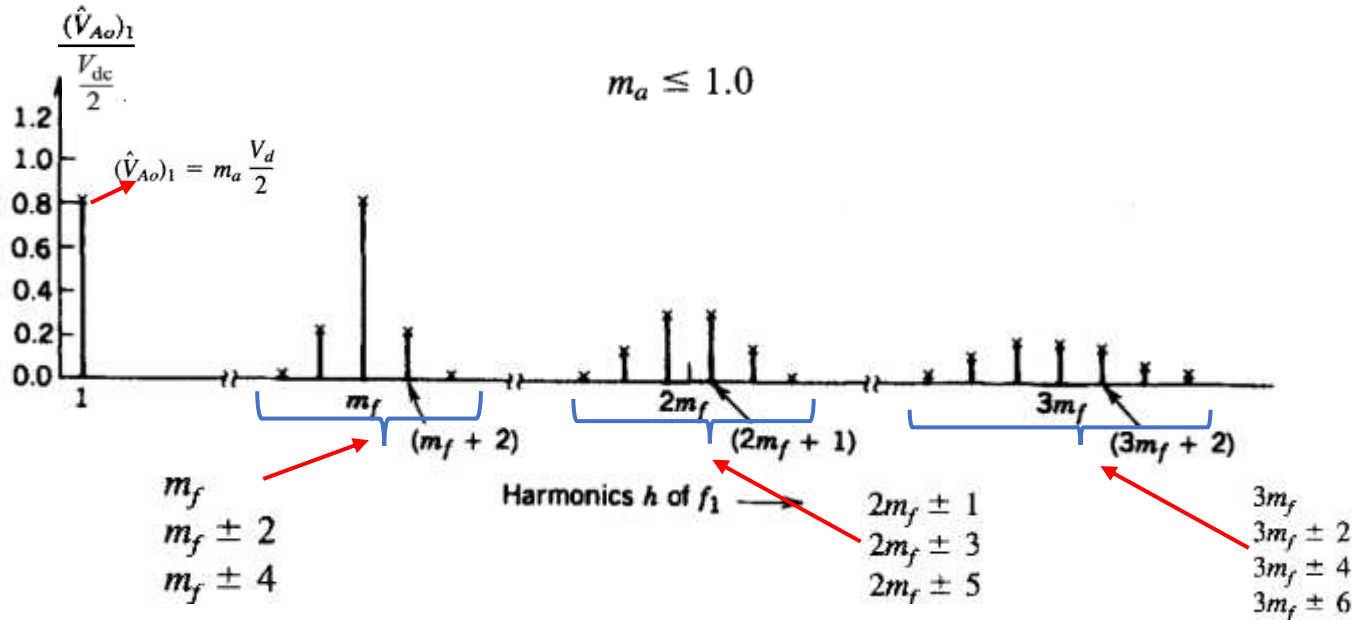
## - Harmonics in PWM Switching



1. The peak amplitude of the fundamental-frequency component  $(\hat{V}_{Ao})_1$  is  $m_a$  times  $\frac{V_d}{2}$ .
2. The amplitude of the fundamental-frequency component of the output voltage varies linearly with  $m_a$  (provided  $m_a \leq 1$ ).
3. The harmonics in the inverter output voltage waveform appear as sidebands, centred around the switching frequency and its multiples, that is, around harmonics  $m_f$ ,  $2 m_f$ ,  $3 m_f$ , and so on.

# Harmonics

## - Harmonics in PWM Switching



4. Theoretically, the frequencies at which voltage harmonics occur can be indicated as:

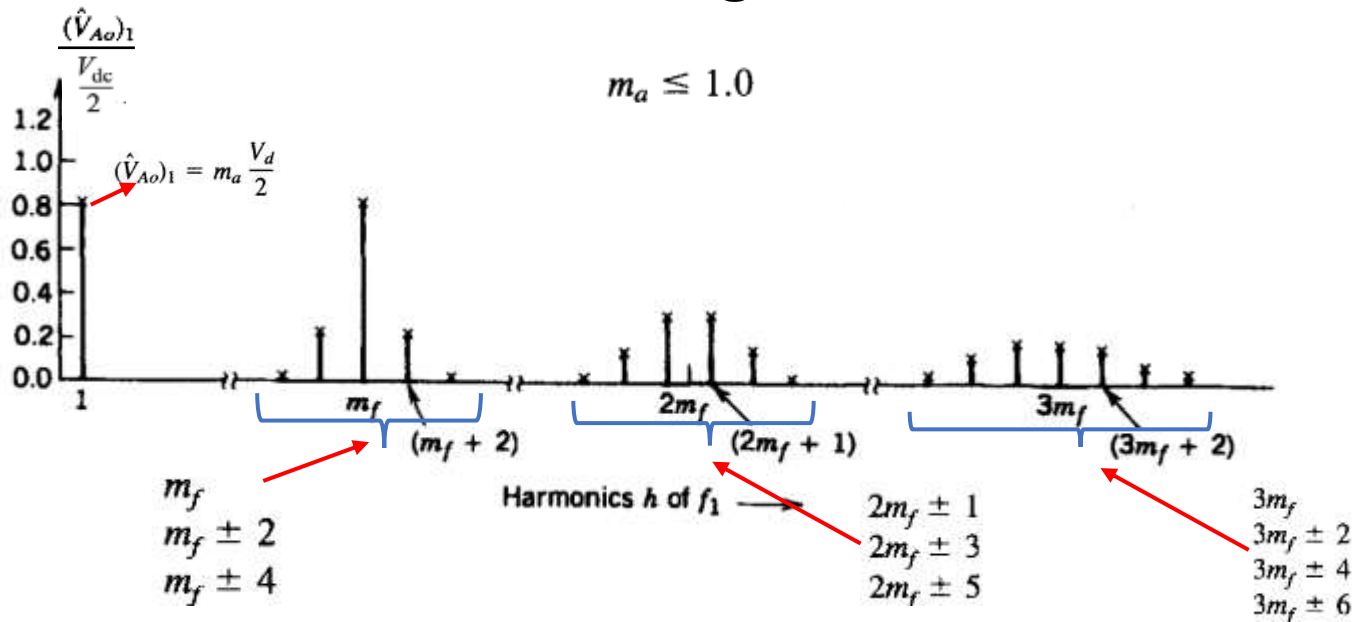
$$f_h = (jm_f \pm k)f_1$$

5. the harmonic order  $h$  corresponds to the  $k^{th}$  sideband of  $j$  times the frequency modulation ratio  $m_f$

$$h = j(m_f) \pm k$$

# Harmonics

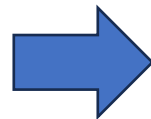
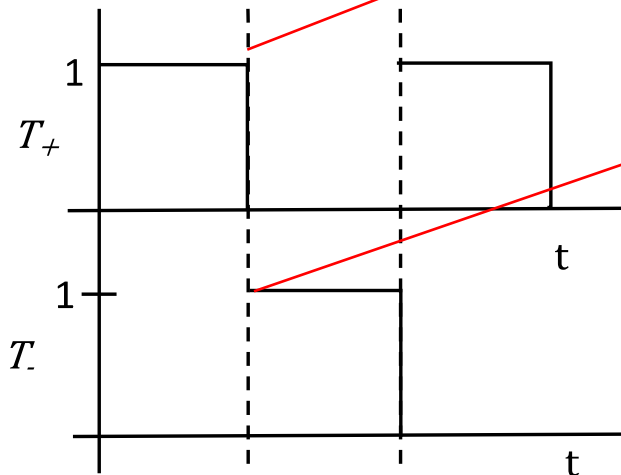
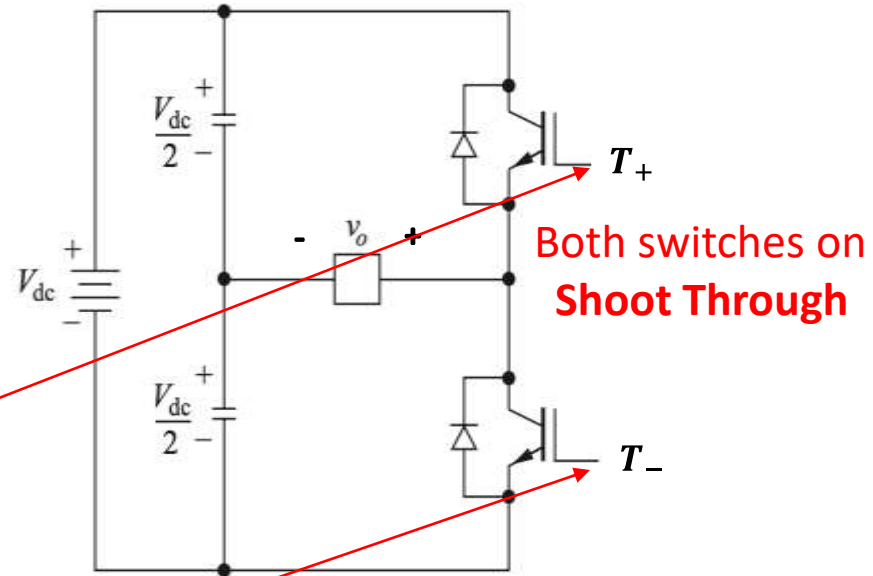
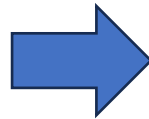
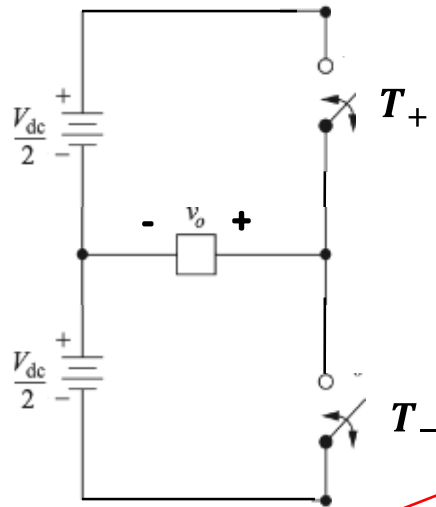
## - Harmonics in PWM Switching



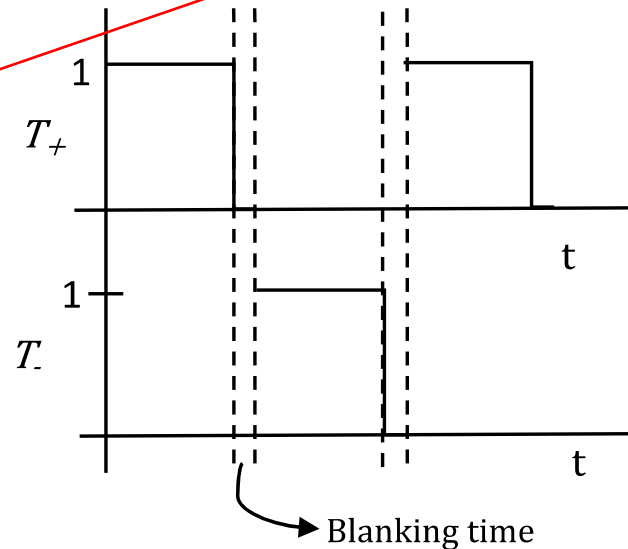
6. For odd values of  $j$ , the harmonics exist only for even values of  $k$ . For even values of  $j$ , the harmonics exist only for odd values of  $k$ .  $f_h = (jm_f \pm k)f_1$

# Practical Half Bridge Inverter

- Made through real components
  - Square Wave Switching

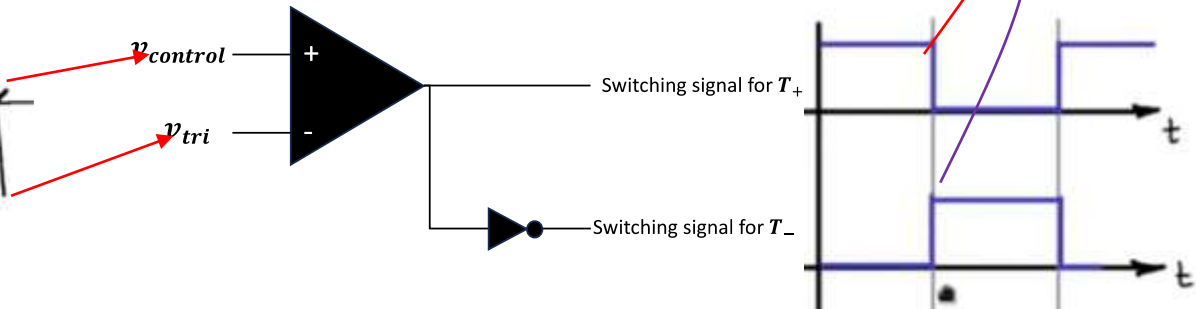
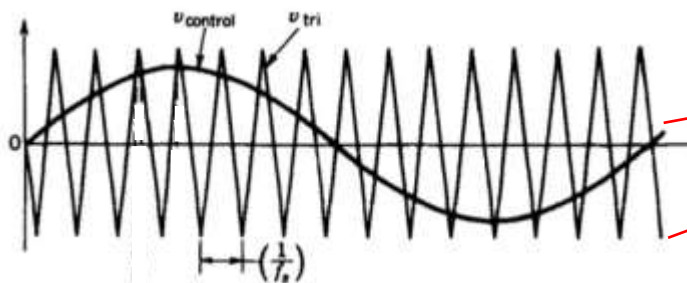
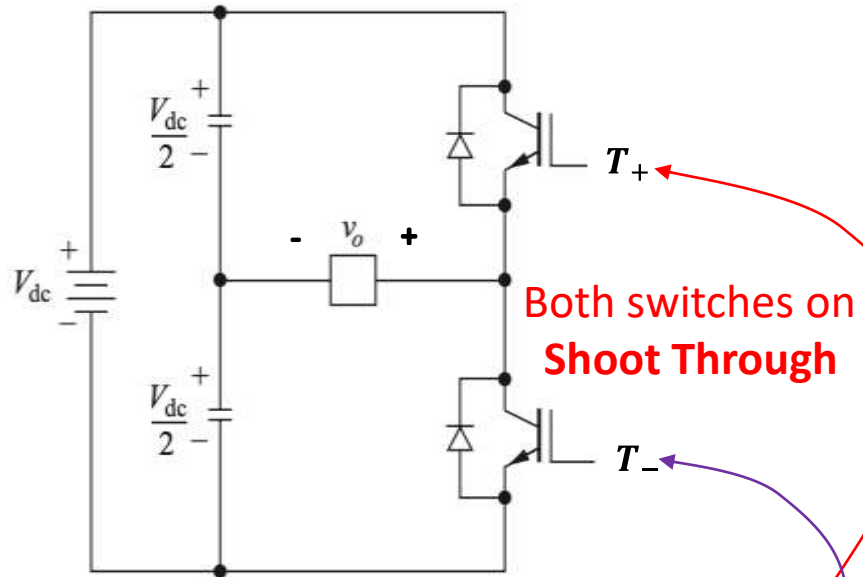
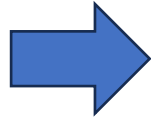
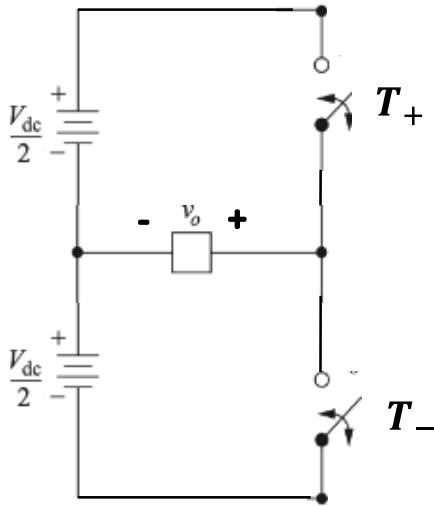


Avoiding Shoot through



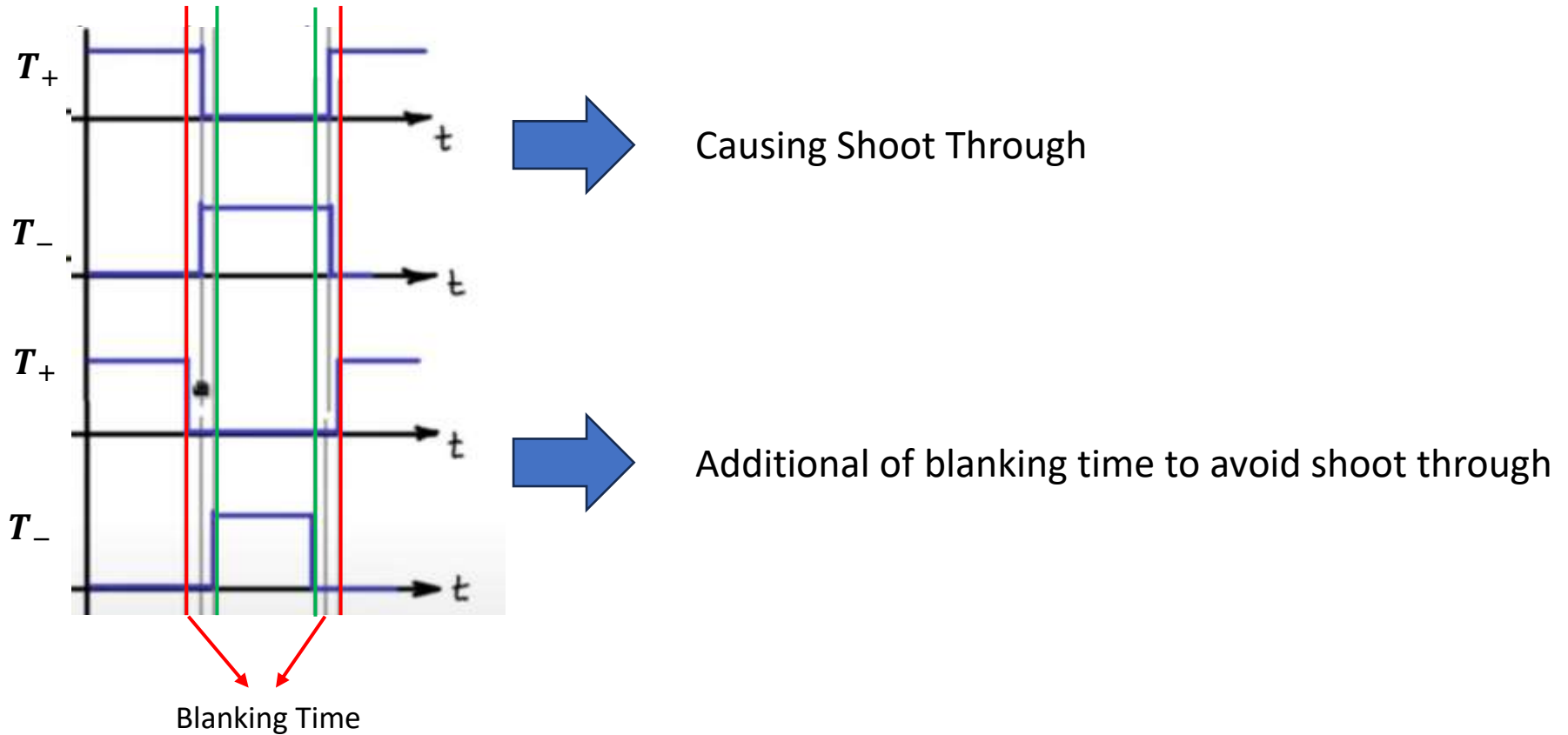
# Practical Half Bridge Inverter

- Made through real components
  - PWM Switching



# Practical Half Bridge Inverter

- Made through real components
  - Avoiding shoot through in PWM Switching



# Avoiding Shoot through

